

INDUSTRIAL ENGINEERING

Administered by the Department of Industrial and Manufacturing Systems Engineering

The undergraduate curriculum in Industrial Engineering leads to the Bachelor of Science degree.

The Industrial Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>, under the commission's "General Criteria and Program Criteria for Industrial and Similarly Named Engineering Programs."

Industrial engineering is a versatile engineering profession that improves processes to help people around the world. Industrial engineers integrate technical expertise with business acumen and effective people skills. They improve processes, organizations, and integrated systems of materials, methods, equipment, information, energy, transportation, and people by reducing waste and increasing quality, safety, and productivity. You will learn how to include the human element in technical problem solving to improve processes in any industry. All industries employ industrial engineers, including, but not limited to, manufacturing, healthcare, entertainment, supply chain, banking, energy, and transportation.

Program Educational Objectives

The Industrial Engineering (IE) Program educates its future graduates to accomplish its program educational objectives (PEOs) in their early careers.

Specifically, the IE Program prepares its majors so that, within a few years after graduation, graduates' attainments are:

- Industrial engineering solutions that are effectively created and communicated and consider relevant stakeholders and ramifications.
- Team goal accomplishment through productive interactions and leadership.
- New capabilities, skills, and knowledge that advance professional practice and enable career advancement.

Student Learning Outcomes

The IE Program currently has the following student outcomes describing what IE majors are expected to know and be able to do by the time of graduation.

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

- An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- An ability to communicate effectively with a range of audiences.
- An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
- An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
- An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

The Industrial Engineering undergraduate curriculum provides students with fundamental knowledge in mathematics and science, engineering science, social science, and humanities as well as professional industrial engineering coursework. In particular, the curriculum includes design, analysis, operation and improvement of integrated systems that produce or supply products or services in an effective, efficient, sustainable and socially responsible manner. A capstone design course provides students with an opportunity to solve open-ended problems with an industrial partner.

Students are strongly encouraged to participate in one or more key activities outside of the regular curriculum to improve their professional skills. Internships and co-op positions in industry provide real-world experiences in the profession and excellent insight on career decisions. Study abroad opportunities, ranging from ten days to a full semester, help students see problems from different perspectives. The department offers many undergraduate research opportunities to work with faculty and graduate students on cutting-edge problems. Other examples of extracurricular opportunities include leadership positions in student organizations and working as a peer mentor or tutor.

Qualified juniors and seniors interested in graduate studies may apply to the Graduate College to concurrently pursue both 1) B.S. and M.S. or M.Eng. degrees in Industrial Engineering, Systems Engineering, or Engineering Management, or 2) B.S. and M.B.A. degrees. For further information, please refer to the Graduate / Graduate College / Concurrent Programs tab.

Curriculum in Industrial Engineering

Administered by the Department of Industrial and Manufacturing Systems Engineering.

Leading to the degree Bachelor of Science.

Total credits required: 123 cr.

See also Basic Program and Special Programs. Grades of C or better are required for any transfer credit course that is applied to the degree program but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA. Note: Department does not allow Pass/Not Pass credits to be used to meet graduation requirements.

International Perspectives: 3 cr.¹

U.S. Cultures and Communities: 3 cr.¹

¹These university requirements will add to the minimum credits of the program unless the university-approved courses are also allowed by the department to meet other course requirements within the degree program.

U.S. cultures and communities and international perspectives courses may not be taken Pass/Not Pass.

Communication Proficiency/Library requirements: 7 CR.:

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1

Remaining Communication courses: 9 cr.

ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or Better in this course)	3
SPCM 2120	Fundamentals of Public Speaking	3
ENGL 3140	Technical Communication	3
Total Credits		9

Social Sciences and Humanities Electives: 12 cr.

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Students must choose SSH courses from at least 2 different programs.

²For Social Sciences and Humanities, and IE Program Electives, choose from the department approved list. (https://www.imse.iastate.edu/files/2023/08/Elective_List_23-24.pdf)

Basic Program: 24 cr.

³

A minimum GPA of 2.00 required for this set of courses (please note that transfer course grades will not be calculated into the Basic Program GPA). See Requirement for Entry into Professional Program in College of Engineering Overview section.

³See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.

CHEM 1670	General Chemistry for Engineering Students	4
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3

ENGR 1010	Engineering Orientation	R
LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
IE 1600	Industrial Engineering Problem Solving with Computer Applications	3

Total Credits **24**

Math and Physical Science: 17 cr.

MATH 2650	Calculus III	4
MATH 2670	Elementary Differential Equations and Laplace Transforms	4
STAT 3031	Probability and Statistical Inference for Engineers	4
PHYS 2320	Introduction to Classical Physics II	4
PHYS 2320L	Introduction to Classical Physics II Laboratory	1

Total Credits **17**

Industrial Engineering Core: 37 cr.

A minimum GPA of 2.00 required for this set of courses (please note that transfer course grades will not be calculated into the Core GPA):

IE 1020	Industrial Engineering Profession	R
IE 2220	Design & Analysis Methods for System Improvements	3
IE 2480	Engineering System Design, Manufacturing Processes	3
IE 2710	Applied Ergonomics and Work Design	3
IE 3050	Engineering Economic Analysis	3
IE 3120	Optimization	3
IE 3410	Production Systems	3
IE 3480	Solidification Processes	3
IE 3610	Statistical Quality Assurance	3
IE 4130	Stochastic Modeling, Analysis and Simulation	4
IE 4320	Industrial Automation	3
IE 4410	Industrial Engineering Design	3
IE 4480	Manufacturing Systems Engineering	3

Total Credits **37**

Other Remaining Courses: 24 cr.

²

CE 2710	Engineering Foundations of Statics	3
& CE 2720	and Applied Engineering Statics	
or CE 2740	Engineering Statics	
MATE 2730	Principles of Materials Engineering	3
ME 2310	Engineering Thermodynamics I	3

IE Program Elective	IE Program Elective	15
Total Credits		24

IE PROGRAM ELECTIVES: 15 cr.

Fifteen credits of IE program electives are required, of which:
 6 must be from IE Courses in Advanced Manufacturing, Operations Research and Analytics, or Human Factors and Ergonomics.
 3 must be from IE Courses in System Engineering or Engineering Management.
 3 must be from department approved list of STEM courses.
 3 from department approved list

Seminar/Co-op/Internships:

Optional co-op/internship courses

See also the following grid showing course template by semester:

4-Year Plan of Study for Industrial Engineering

Industrial Engineering, B.S.

First Year

Fall	Credits Spring	Credits
SSH Elective	3 SSH Elective	3
IE 1600	3 MATH 1660	4
MATH 1650	4 ENGL 1500	3
CHEM 1670	4 LIB 1600	1
ENGR 1010	R IE 1020	R
	PHYS 2310	4
	PHYS 2310L	1
	14	16

Second Year

Fall	Credits Spring	Credits
MATH 2650	4 MATH 2670	4
IE 2480	3 IE 2220	3
MATE 2730	3 IE 2710	3
PHYS 2320	4 STAT 3031	4
PHYS 2320L	1	
ENGL 2500	3	
	18	14

Third Year

Fall	Credits Spring	Credits
IE 3050	3 IE 3480	3
IE 3410	3 IE 3610	3
IE 3120	3 CE 2720	2

SPCM 2120	3 IE Program Elective	3
IE 4320	3 SSH Elective	3
CE 2710	1	
	16	14

Fourth Year

Fall	Credits Spring	Credits
IE 4130	4 IE 4410	3
ME 2310	3 IE 4480	3
IE Program Elective	3 IE Program Elective	3
ENGL 3140	3 IE Program Elective	3
SSH Elective	3 IE Program Elective	3
	16	15

Engineering Sales Minor

The Engineering Sales Minor is multidisciplinary and open to undergraduates in the College of Engineering. The minor requires 15 credits, including at least 6 credits in courses numbered 3000 or above taken at Iowa State University. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement.

IE 4500	Technical Sales for Engineers	3
IE 4300	Entrepreneurial Product Engineering	3
MKT 3400	Principles of Marketing	3
MKT 4500	Advanced Professional Selling	3
	And one of the following:	3
IE 3050	Engineering Economic Analysis	
CE 2060	Engineering Economic Analysis and Professional Issues in Civil Engineering	
Total Credits		15

Concurrent Degrees

If you are considering going to graduate school, the concurrent bachelor's-master's in industrial engineering is an excellent opportunity to obtain both degrees within five years. In this program six credits of graduate course work can be used to satisfy your undergraduate program requirements. After completing the B.S. program, you will be a full-time graduate student for the remainder of the master's program. The following concurrent bachelor's-master's degree program combinations are available to qualified students:

- B.S. IE – M.S. IE (thesis or creative component required)
- B.S. IE – M.Eng. IE (coursework-only)
- B.S. IE – M.Eng. Systems Engineering (coursework-only)
- B.S. IE – M.Eng. Engineering Management (coursework-only)

Program Policies

1. Up to two semesters of concurrent enrollment are allowed.
2. Students can enroll in up to twelve credits of coursework at the 5000-level in industrial engineering for both the B.S. and M.S./M.Eng. degrees. Six of the shared credits must be taken before being classified as a concurrent student.
3. Students must take at least three credits of 5000-level courses each semester during concurrent enrollment and pay graduate tuition and fees.
4. Students participating in the concurrent program must adhere to the requirements for the graduate degree program in which they are enrolled.

Admission Requirements

Students must have a record of high academic achievement and should have at least a 3.00. Students must be within 30 credits of completing the requirements for the B.S. degree before applying to the concurrent program. Prospective students **must** speak with their undergraduate advisor prior to applying.

Concurrent B.S. in Industrial Engineering and MBA in the College of Business

Accelerate Your Career with an MBA

Today's successful engineers often need strong business management skills to complement their degree in engineering. These skills are especially important for engineers who are interested in working for technical companies, launching their own business ventures, or pursuing nontraditional jobs such as consulting. The MBA provides an opportunity for competitive advantage for advancement into management positions.

The Colleges of Business and Engineering offer an integrated, concurrent program leading to Bachelor of Science and Master of Business Administration degrees. The program is designed to be completed in a minimum of five academic years or ten semesters.

The actual time for completion may be longer depending upon semester course loads, internships, and course sequencing. Careful planning with an academic advisor is essential throughout the program. This program is slightly accelerated through the first six semesters to complete the general education requirements. The remaining four semesters integrate the B.S. in IE requirements and the MBA courses. Twelve credits will be shared in common and will fulfill the requirements for both the B.S. in IE and the MBA.

Summer study is possible, although students are strongly encouraged to complete at least two internships or co-ops in engineering and business for practical experience during their academic program.

Preparation for the B.S. IE/MBA

Students interested in pursuing the MBA need to demonstrate exemplary academic performance throughout their undergraduate program. Mastery of communications, economic theory, global and diversity perspectives, statistical data analysis, and the core professional curriculum is of particular importance.

The Selection Process

The program is demanding, and admission is selective. Students are expected to exhibit superior intellectual ability, strong leadership attributes, and must be highly motivated and career focused. Although not absolute minimums, admitted students typically will have earned a cumulative GPA of 3.00 or above and have scored 600 or higher on the Graduate Management Admissions Test (GMAT). All application materials, however, are taken into account collectively in the applicant review and student selection process.

GRADUATE STUDY

The department offers three industrial engineering graduate degrees: Master of Engineering (M.Eng.), Master of Science (M.S.), and Doctor of Philosophy (Ph.D.). A graduate minor is available to graduate students having a major in another department. The M.Eng. degree consists of coursework designed to improve professional expertise in industrial engineering. The M.S. and Ph.D. degrees are designed to improve the student's capability to conduct research as well as advancing their professional expertise. In conjunction with the Department of Mechanical Engineering, the department offers a certificate in advanced manufacturing.

To be considered for admission into the graduate program, the applicant should have a bachelor's degree in industrial engineering or related field from a college, university, or technical school of recognized standing. Other educational backgrounds will be considered on an individual basis. High academic achievement or other persuasive evidence of professional accomplishments is expected for admission to the program. An applicant may apply as a direct-entry Ph.D. applicant when they have not earned a master's degree.

The graduate program offers advanced study in advanced manufacturing, ergonomics/human factors, operations research/analytics, systems engineering and engineering management.

Well-qualified juniors and seniors in industrial engineering who are interested in graduate study may apply for concurrent enrollment to simultaneously pursue both the industrial engineering bachelor's degree and an M.Eng. or M.S. degree. Two other attractive concurrent degree options are the industrial engineering bachelor's degree concurrent with a M.Eng. Engineering Management degree or an M.Eng. Systems Engineering degree. For those more interesting in business management,

we also offer a concurrent Master of Business Administration degree from the business college. For additional information about graduate degree programs, admission criteria, and procedures refer to <https://www.imse.iastate.edu/graduate-program/>.